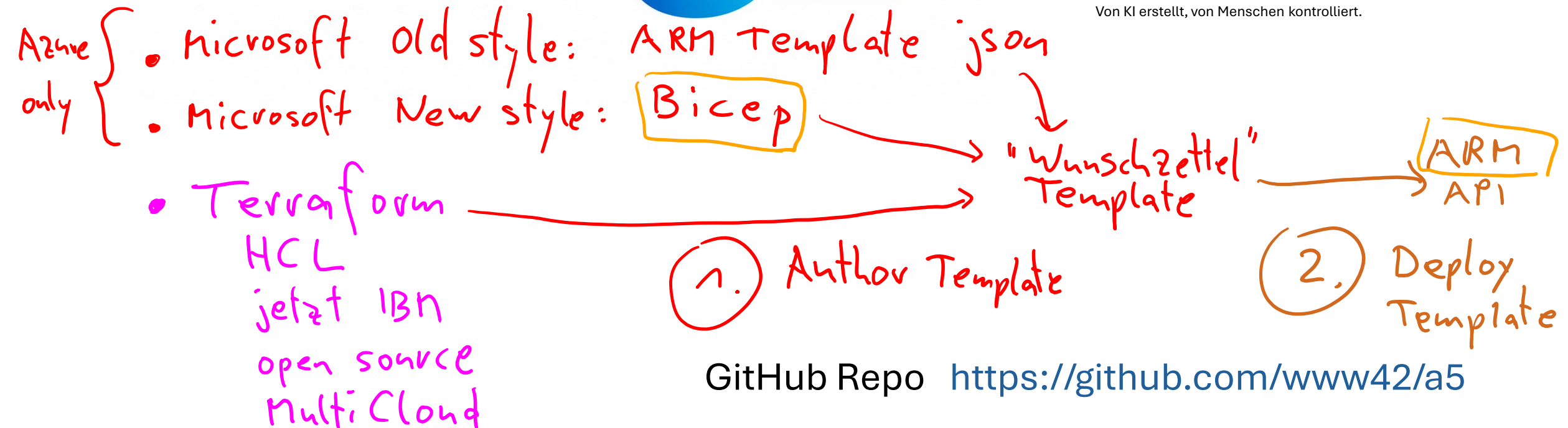




30.06.2026
Tag 7

Von KI erstellt, von Menschen kontrolliert.



AZ-104

Exam AZ-104
700 i
620 i
700

Jeff Shover
Inventor PowerShell
1) Get-Command
2) Get-Help -Online
3) Get-Member

Learning Paths:

Daily MeasureUp

Vormittag
Glückswad
Thema

1. → Manage identities and governance in Azure
2. → Prerequisites for Azure administrators – CloudShell and templates
- Implement and manage storage in Azure
- Configure and manage virtual networks for Azure administrators
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

1. Woche ✓

2. Woche Freitag 03.07.

ich bin nicht da 10:45 - 13:00
ich bin in Mannheim ab 13:00

3. Woche

ich bin in Heidelberg oder Mannheim

4. Woche 13.07. - 17.07. Neuer Trainer Online

5. Woche Prüfungsvorbereitung
Prüfung

ich bin wieder da

What happened yesterday



	Power shell shell	Bash shell
Power shell cmdlets	✓	✗
Azure CLI az....	✓	✓

az group list -o table

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➔ Prerequisites for Azure administrators – CloudShell and templates



① Author Template



Deploy Azure resources with templates

② Deploy Template

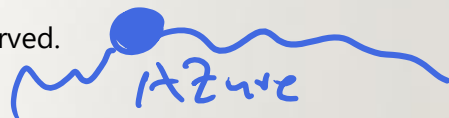
Bicep (New style)

↓ Unsichtbar

ARM json (old style)

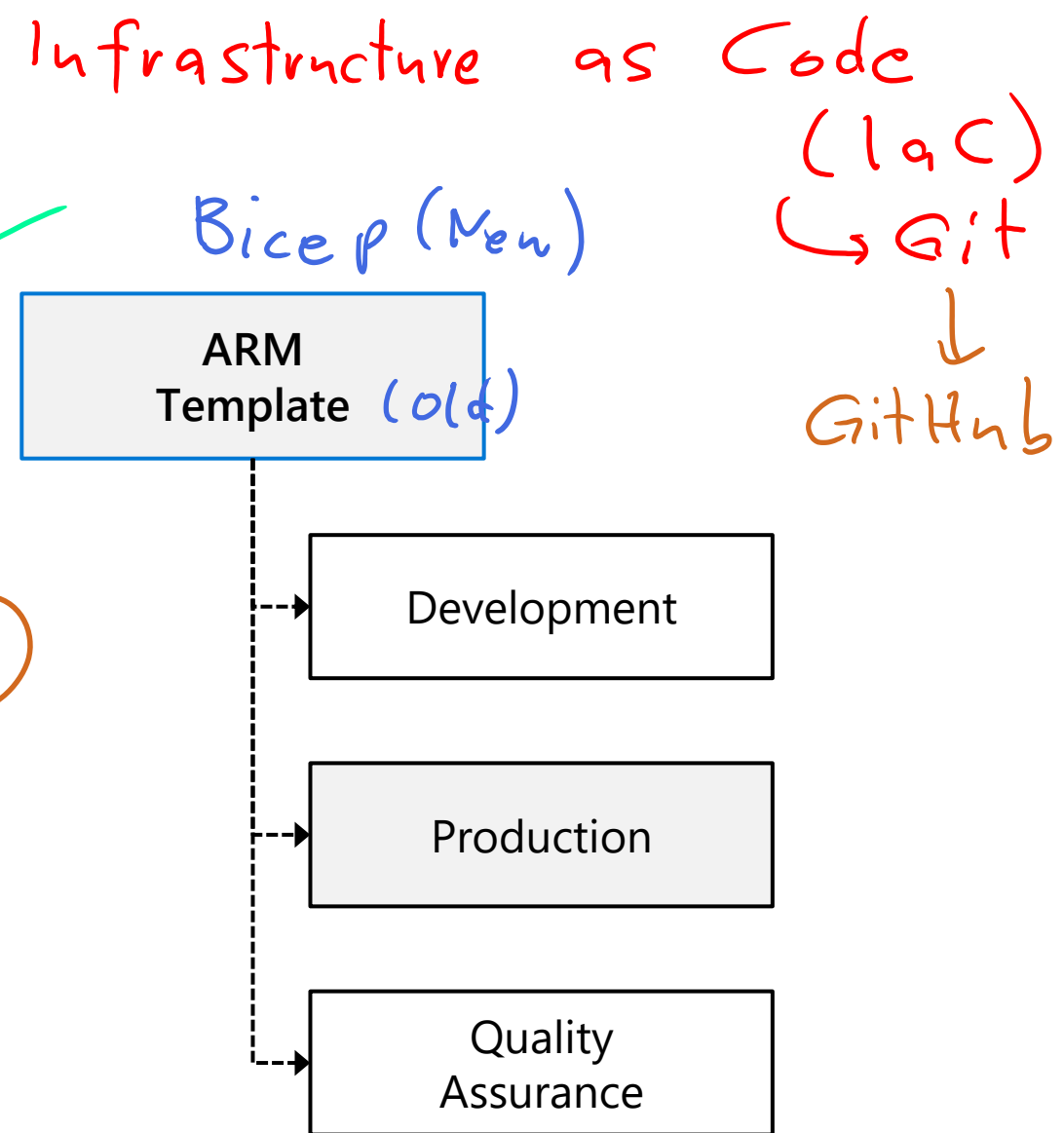


ARM API



Review ARM Template Advantages

- Improves consistency and promotes reuse
- Reduce manual, error prone, and repetitive tasks
- Express complex deployments
- Express requirements through code
- Provides validation tasks
- Modular and can be linked
- Simplifies orchestration



Explore the JSON Template Schema

- Defines all the Resource manager resources in a deployment
- Written in JSON
- A collection of key-value pairs
- Each key is a string
- Each value can be a string, number, Boolean expression, list of values, object

Get-NetIPAddress

Old Style

```
{
  "$schema": "http://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "",
  "parameters": {},
  "variables": {},
  "functions": [],
  "resources": [],
  "outputs": {}
}
```

Annotations for the JSON Template Schema:

- Key**: Points to the key part of a key-value pair (e.g., "\$schema").
- Gänsefüßchen (Double Quotes)**: Points to the double quotes around the key and value.
- Tüddelchen**: Points to the colon separator between the key and value.
- Doppelpunkt (Colon)**: Points to the colon separator between the key and value.
- Value**: Points to the value part of a key-value pair (e.g., "http://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#").
- Comma**: Points to the comma separator between key-value pairs.
- List**: Points to the array notation [] used for "functions" and "resources".
- Object**: Points to the object notation {} used for "parameters", "variables", and "outputs".
- Single Quote**: Points to the single quote character used in the example 'foo'.

Resources

VS Code → Lang Server json

'foo'

Explore the JSON Template Parameters

template.json
parameters.json

- Specifies which values are configurable when the template runs
- This example has two parameters: one for a VM's username (adminUsername), and one for its password (adminPassword)

```
"parameters": {  
  "adminUsername": {  
    "type": "string",  
    "metadata": {  
      "description": "Username for the VM."  
    }  
  },  
  "adminPassword": {  
    "type": "securestring",  
    "metadata": {  
      "description": "Password for the VM."  
    }  
  }  
}
```

Beispiel Value:
LocalAdmin
~~Admin~~

Pa55w.rd123

Consider Azure Bicep Files

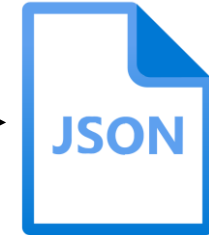
- Simpler syntax for writing templates
- Smaller module files you can reference from a main template
- Automatically detect dependencies between your resources
- Visual Studio Code extension with validation and IntelliSense

No Double Quotes
No Comma
Symbolic Name

Bicep file

```
resource storageAccount
'Microsoft.Storage/storageAccounts@
2021-01-01' = {
  name: storageAccountName
  location: location
  tags: {
    displayName: storageAccountName
  }
  kind: 'StorageV2'
  sku: {
    name: 'Standard_LRS'
  }
}
```

Name



output: storageAccount.URL

Most Benefit: Modules!

1. Author Template

- Portal : Reverse Templating (Export)
- Quick start Templates
- From Scratch (vs code + Bicep Extension)
- Lass Dir Vorschläge unterbreiten!

2. Deploy Template Resource Group

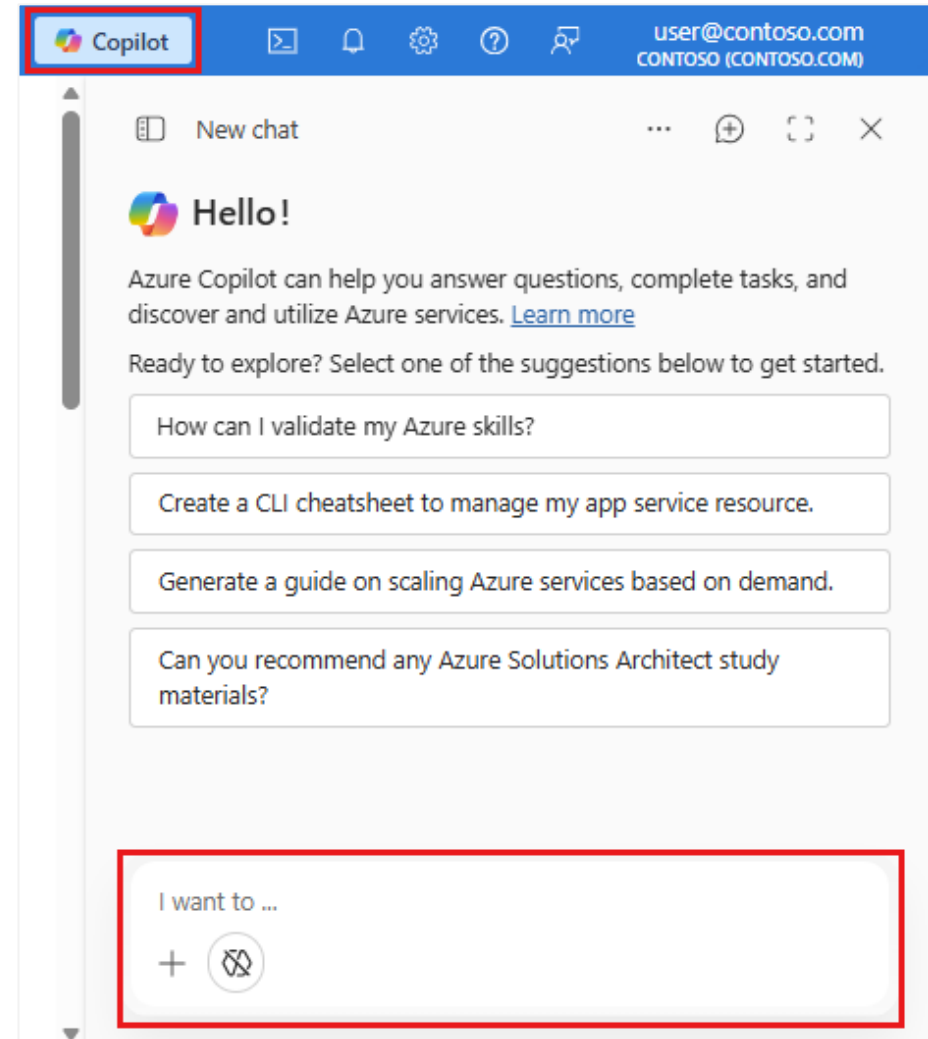
- Azure Portal : Custom Deployment Old style only
- Powershell : New-AzResourceGroupDeployment
-Templatefile ... bicep
- Azure CLI : az deployment group ...

And now for something
completely different...



Microsoft Copilot for Azure

- Access in the Azure portal, through the Azure mobile app, or through AI Shell
- Helps you design, operate, optimize, and troubleshoot your Azure apps and infrastructure
- Performs many basic tasks
 - Deploy and manage virtual machines
 - Design, troubleshoot, and secure networks
 - Generate PowerShell and Azure CLI scripts
 - Analyze, estimate, and optimize costs
 - Discover and deploy workload templates



Hente Nachmittag:

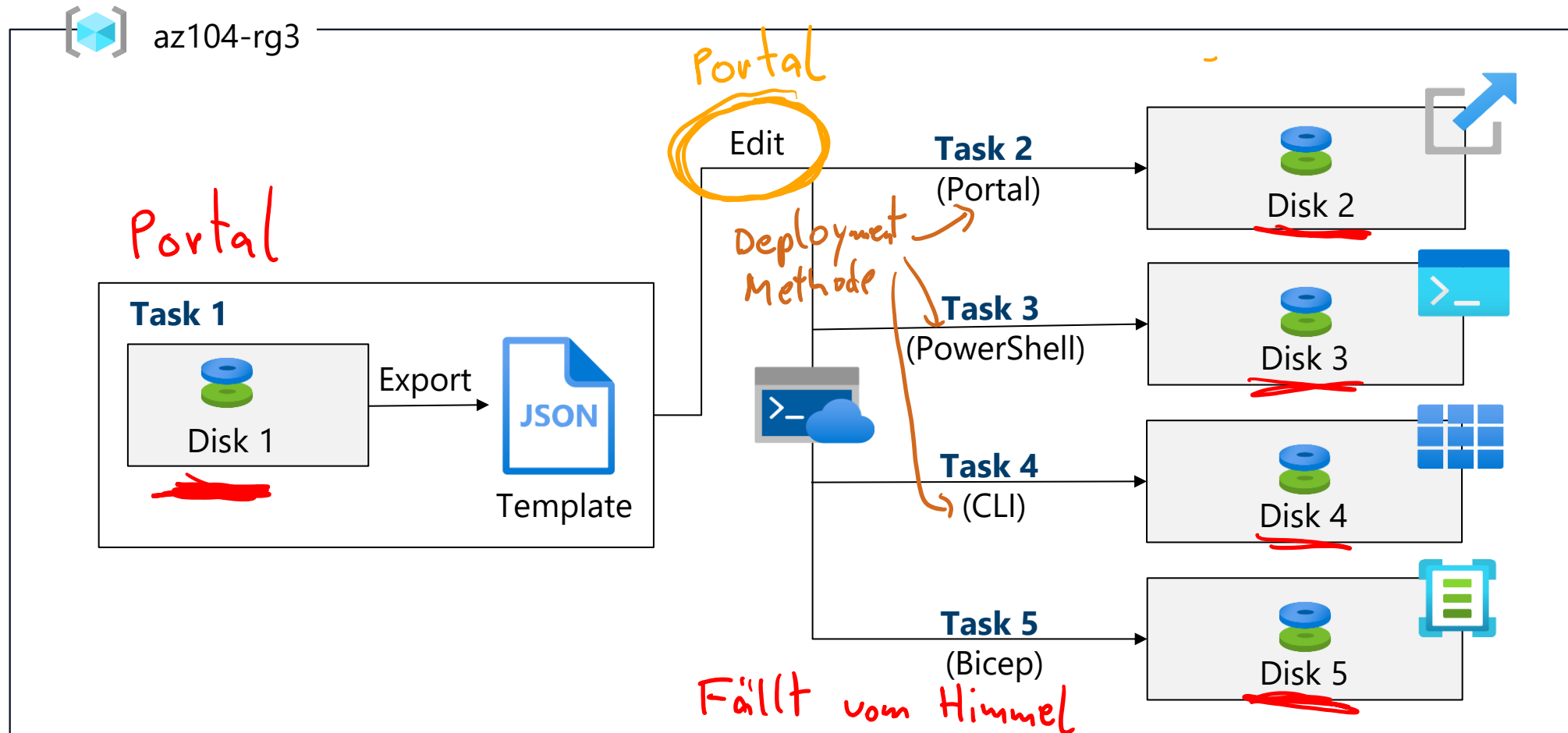
AZ-104

Lab 03 - Manage Azure resources by Using ARM Templates



Lab 03 – Architecture diagram

VS Code
Shortcuts
Search + Replace `ctrl-H`



Labs



End of presentation



